SECURITY · MACHINE LEARNING · CRYPTOGRAPHY

Adaptive security for cloud sessions.

An educational prototype exploring identity, encryption and explainable session risk.

SYNTHETIC DATA

EDUCATIONAL PROTOTYPE

MFA + AES-256-GCM + ML



THE IDEA

One console. Multiple layers.

A practical workspace for exploring cloud-session security.

IDENTITY RISK ENCRYPTION AUDIT

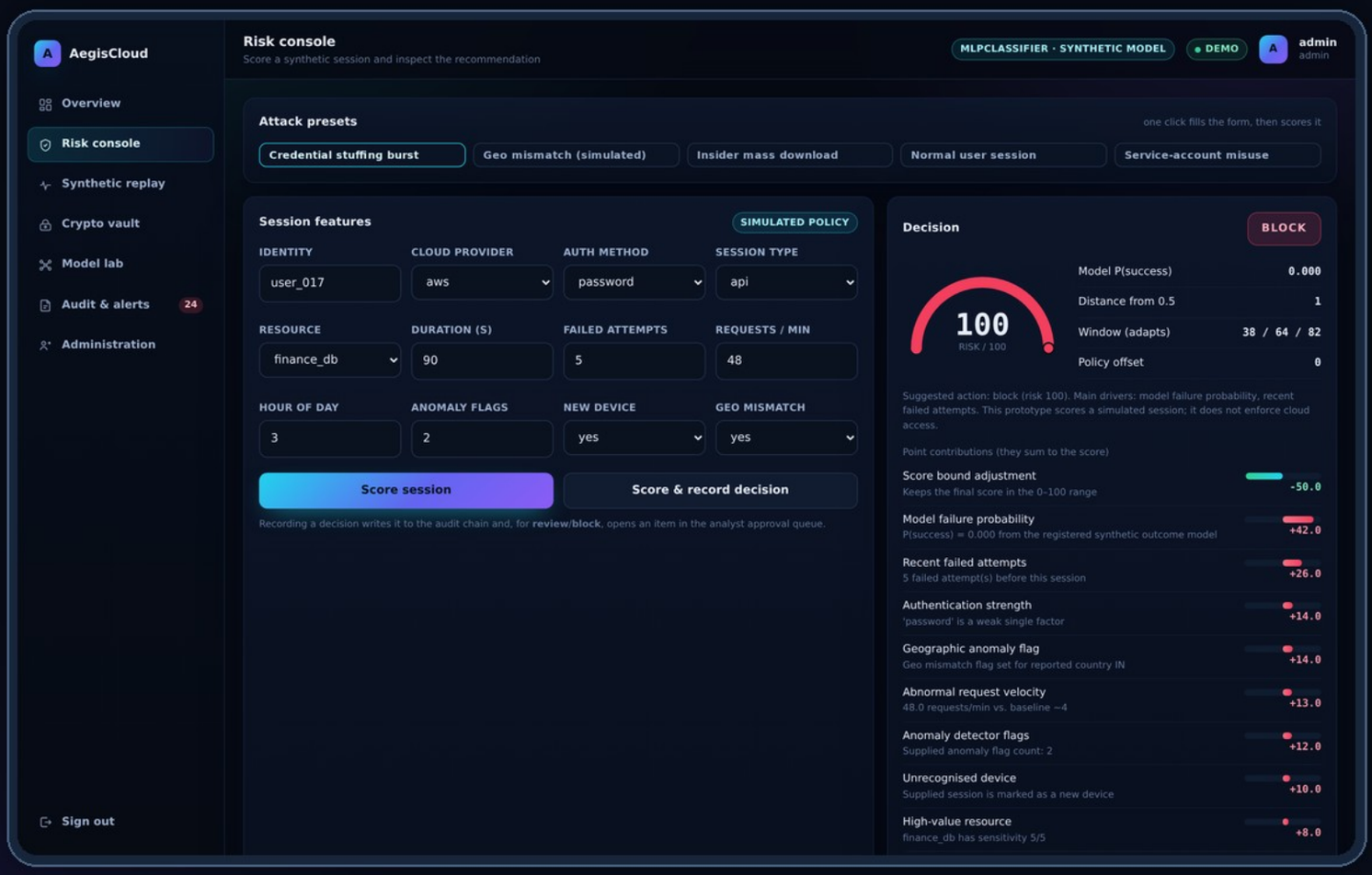


Actual AegisCloud dashboard · fictional demo data

EXPLAINABLE RISK

Risk you can explain.

Model output + behavioral signals become a clear 0–100 score with additive contributions.



ALLOW / STEP-UP / REVIEW / BLOCK

Recommendations only; no cloud access is enforced.

SYNTHETIC EXPERIMENT

Measured results. Honest boundaries.

MLPClassifier · 1,350 synthetic holdout sessions

ACCURACY	SUCCESS F1	ROC-AUC
83.04%	0.8829	0.8701
synthetic holdout	success = class 1	synthetic outcomes

TRAINING DESIGN

9,000 generated sessions

70% training → 15% validation → 15% holdout

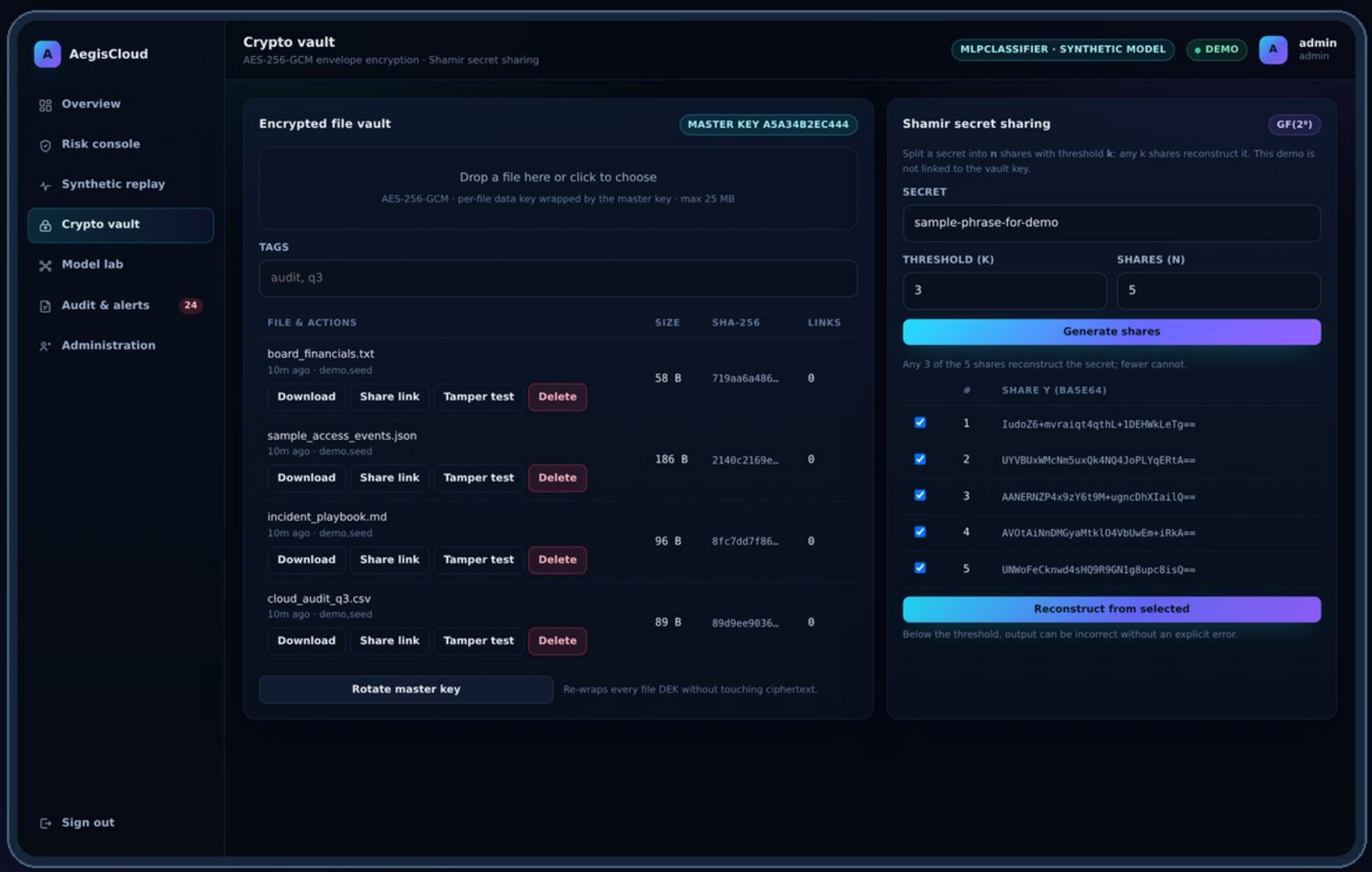
Success is the positive class — not an attack label.
26 isolated tests passed. LSTM/CNN training was not run.

DATA PROTECTION

Security is more than a score.

Authenticated file encryption, tamper checks and a hash-linked audit trail.

AES-256-GCM · TAMPER TEST · AUDIT



Secret-sharing is a separate demo; local keys are not a production KMS.

LEARN / BUILD / IMPROVE

Built to learn. Ready to explore.

A prototype for asking better security questions — not a production security product.

NEXT UP

- Evaluate with real, consented, labelled telemetry
- Integrate provider-side access enforcement
- Add secure MFA delivery and managed keys
- Run the LSTM / 1D-CNN experiment

EXPLORE THE PROJECT

github.com/joyboyzz/AegisCloud

Feedback and ideas are welcome.